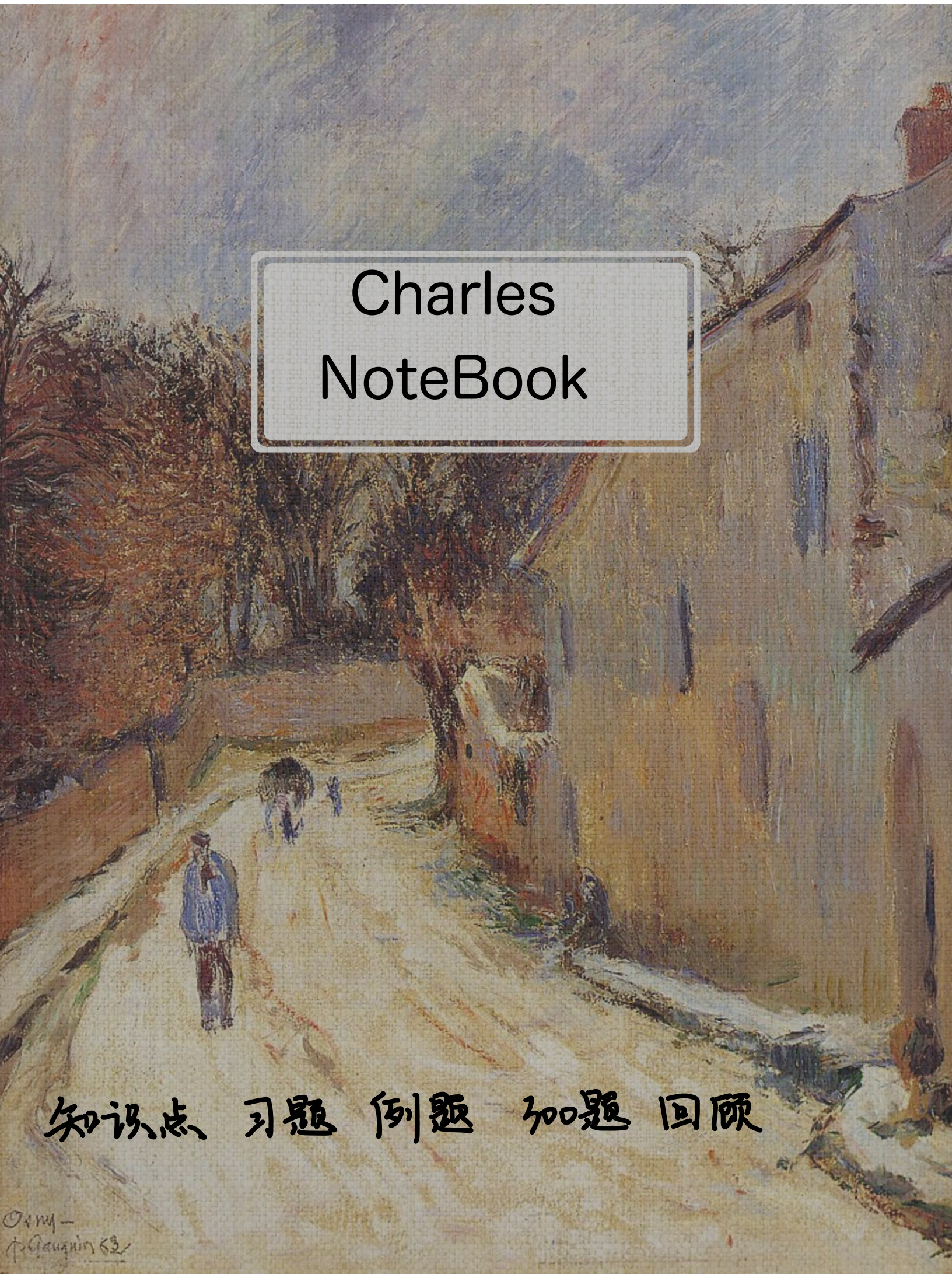


An impressionist painting of a street scene. In the foreground, a man in a blue coat and hat stands on a dirt path. To his right, a large, leafy tree stands next to a tall, light-colored building. In the background, more trees and a few other figures are visible. The sky is a mix of blue and white, suggesting a bright day. The overall style is loose and painterly, with visible brushstrokes.

Charles NoteBook

知识点 习题 例题 加题 回顾

一元函数微分学的应用

物理应用 $v = \frac{ds}{dt}$ $a = \frac{dv}{dt}$ “ $\frac{dA}{dB}$ 表示 A 对 B 的变化率” $\star a = \frac{dv}{dt} = \frac{dv}{ds} \frac{ds}{dt}$

相关变化率 便可求 $\left(\frac{dy}{dt}\right) = \frac{dy}{dx} \frac{dx}{dt} = \underbrace{f'(x)}_{\text{相关变化率 (往往已知)}} \cdot \underbrace{\left(\frac{dx}{dt}\right)}_{\text{另告之}}$

几何应用 曲率公式 $k = \frac{|y''|}{[1+(y')^2]^{\frac{3}{2}}}$ 形式 $\bar{x} = \frac{\iint x d\sigma}{\iint d\sigma} = \frac{\int_a^b dx \int_0^{f(x)} y dy}{\int_a^b dx \int_0^{f(x)} dy} = \frac{\int_a^b x f(x) dx}{\int_a^b f(x) dx}$

曲率半径 $R = \frac{1}{k} = \frac{[1+(y')^2]^{\frac{3}{2}}}{|y''|} \quad (y'' \neq 0)$ $\bar{y} = \frac{\iint y d\sigma}{\iint d\sigma} = \frac{\int_a^b dx \int_0^{f(x)} y dy}{\int_a^b dx \int_0^{f(x)} dy} = \frac{\frac{1}{2} \int_a^b f(x)^2 dx}{\int_a^b f(x) dx}$

弧长 直角坐标 $S = \int_a^b \sqrt{1+(y')^2} dx$

参数方程 $S = \int_a^b \sqrt{(x')^2 + (y')^2} dt$

极坐标系 $S = \int_\alpha^\beta \sqrt{(r\theta')^2 + (r\theta)^2} d\theta$

表面积 $y = y(x)$ $S = 2\pi \int_a^b |y(x)| \sqrt{1+(y')^2} dx$

参数方程 $S = 2\pi \int_a^b |y(t)| \sqrt{(x')^2 + (y')^2} dt$

$ds = \sqrt{(dx)^2 + (dy)^2}$

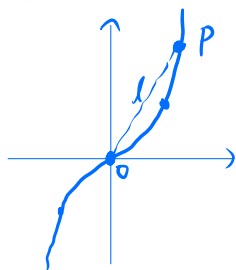
(4) 平行截面面积为已知的立体体积.

在区间 $[a, b]$ 上, 垂直于 x 轴的平面截立体 Ω 所得到的截面面积为 x 的连续函数 $A(x)$, 则 Ω 的体积为

$$V = \int_a^b A(x) dx.$$

例 15.1

$\star$ 例 15.1 已知动点 P 在曲线 $y = x^3$ 上运动, 记坐标原点与点 P 间的距离为 l . 若点 P 的横坐标对时间的变化率为常数 v_0 , 则当点 P 运动到点 $(1, 1)$ 时, l 对时间的变化率是_____.



$$l = \sqrt{x^2 + y^2}$$

$$\frac{dx}{dt} = v_0 \quad \frac{dl}{dt} = \frac{dl}{dx} \frac{dx}{dt} = \frac{1}{2} (x^2 + y^2)^{-\frac{1}{2}} (2x + 6x^5) \cdot v_0$$

$$x=1, y=1 \text{ 时 } \frac{dl}{dt} = 2\sqrt{2} v_0$$

例 15.2

例 15.2 曲线 $\begin{cases} x = \cos^3 t, \\ y = \sin^3 t \end{cases}$ 在 $t = \frac{\pi}{4}$ 对应点处的曲率为_____.

$$y' = \frac{dy/dt}{dx/dt} = \frac{3\sin^2 t \cdot (\cos t)}{3\cos^2 t \cdot (-\sin t)} = \frac{\sin t}{-\cos t} = -\tan t$$

$$y'' = \frac{dy'/dt}{dx/dt} = \frac{\frac{1}{\cos^2 t}}{3\cos^2 t \cdot (-\sin t)} = \frac{-1}{3\cos^4 t \cdot \sin t}$$

$$k = \frac{|y''|}{(1+y'^2)^{\frac{3}{2}}} = \frac{\left| \frac{-1}{3\cos^4 t \cdot \sin t} \right|}{\left(1 + \frac{\sin^2 t}{\cos^2 t} \right)^{\frac{3}{2}}} \bigg|_{t=\frac{\pi}{4}} = \frac{2}{3}$$

一元积分学的应用

物理应用

变力沿直线做功

力: $F(x)$

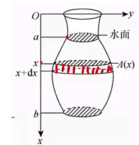
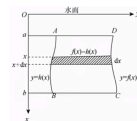
功: $W = \int_a^b F(x) dx$, $dW = F(x) dx$

抽水做功

$W = \rho g \int_a^b x \cdot A(x) dx$

水压力

$P = \rho g \int_a^b x [f(x) - h(x)] dx$



几何应用

平面上的曲边梯形的形心坐标公式

平面曲线的弧长

旋转曲面的表面积

平行截面面积为已知的立体体积

例 15.3

例 15.3 设地球的质量为 M , 半径为 R , 将地面上的质量为 m 的物体举高 H 米, 求克服重力所做的功.

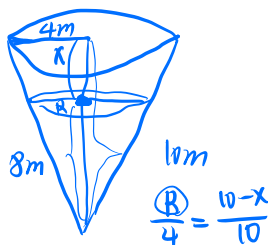
$$\text{重力 } F(x) = \frac{GMm}{x^2}$$

$$W = \int_R^{R+H} \frac{GMm}{x^2} dx$$



例 15.4

例 15.4 有一倒圆锥形容器, 高 10 m, 上底半径 4 m, 水面高 8 m. 求将容器中的水全部从容器顶部抽出所做的功 (水的密度为 1000 kg/m^3 , 重力加速度 g 为 9.8 m/s^2).



$$W = \rho g \int_2^{10} \pi \frac{(10-x)^2}{100} \cdot dx$$

例 15.5

例 15.5 洒水车上的水箱是一个横放的椭圆柱体, 椭圆的尺寸如图 15-15 所示. 当水箱装满水时, 计算水箱的一个端面所受的压力 (水的密度为 1000 kg/m^3 , 重力加速度 g 为 9.8 m/s^2).

$$y = \sqrt{1 - \frac{x^2}{0.75^2}}$$

$$y = -\sqrt{1 - \frac{x^2}{0.75^2}}$$

$$P = \rho g \int_{-0.75}^{0.75} (0.75+x) 2 \sqrt{1 - \frac{x^2}{0.75^2}} dx$$

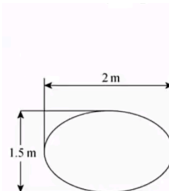


图 15-15

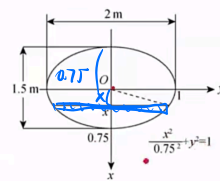
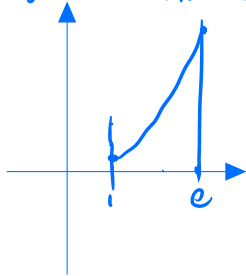


图 15-16

例 15.6

例 15.6 设曲线 L 的方程为 $y = \frac{1}{4}x^2 - \frac{1}{2}\ln x, 1 \leq x \leq e$, D 是由曲线 L , 直线 $x=1, x=e$ 及 x 轴围成的平面图形, 求 D 的形心的横坐标.

$y' = \frac{1}{2}x - \frac{1}{2x} = \frac{1}{2}(x - \frac{1}{x}) \quad x \in [1, e] \Rightarrow y' > 0$



$$\begin{aligned}
 \bar{x} &= \frac{\iint_D x d\sigma}{\iint_D d\sigma} = \frac{\int_1^e dx \int_0^{f(x)} x dy}{\int_1^e f(x) dx} = \frac{\int_1^e x f(x) dx}{\int_1^e f(x) dx} \\
 &= \frac{\int_1^e (\frac{x^3}{4} - \frac{x}{2} \ln x) dx}{\int_1^e (\frac{x^2}{4} - \frac{1}{2} \ln x) dx} = \frac{\frac{x^4}{16} - \frac{1}{2}(\frac{x^2}{2} \ln x - \frac{x^2}{4}) \Big|_1^e}{\frac{x^3}{12} - \frac{1}{2} \ln x + \frac{x}{2} \Big|_1^e} \\
 &= \frac{\frac{e^4}{16} - \frac{1}{2}(\frac{e^2}{2} - \frac{e^2}{4}) - (1 - \frac{1}{2}(-\frac{1}{4}))}{(\frac{e^3}{12} - \frac{e}{2} + \frac{e}{2}) - (\frac{1}{12} + \frac{1}{2})} = \frac{\frac{e^4}{16} - \frac{e^2}{8} + \frac{3}{16}}{\frac{1}{12}e^3 - \frac{7}{12}} \\
 &= \frac{(e^2+1)(e^2-3)/16}{(e^3-7)/12} = \frac{3(e^2+1)(e^2-3)}{4(e^3-7)}
 \end{aligned}$$

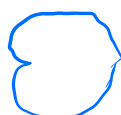
例 15.7

例 15.7 计算曲线 $y = \ln(1-x^2)$ 上相应于 $0 \leq x \leq \frac{1}{2}$ 的一段弧的长度.

$$\begin{aligned}
 S &= \int_0^{\frac{1}{2}} \sqrt{1 + (y')^2} dx = \int_0^{\frac{1}{2}} \sqrt{1 + \frac{4x^2}{(1-x^2)^2}} dx = \int_0^{\frac{1}{2}} \sqrt{\frac{(1+x^2)^2}{(1-x^2)^2}} dx \\
 &= \int_0^{\frac{1}{2}} \frac{1+x^2}{1-x^2} dx = 2 \int_0^{\frac{1}{2}} \frac{dx}{1-x^2} - \int_0^{\frac{1}{2}} 1 dx \stackrel{x=\sin t}{=} -\frac{1}{2} + 2 \int_0^{\frac{\pi}{6}} \frac{dt}{\cos t} \\
 &= -\frac{1}{2} + [2 \ln |\sec t + \tan t|]_0^{\frac{\pi}{6}} = -\frac{1}{2} + \ln 3
 \end{aligned}$$

例 15.8

例 15.8 求心形线 $r = a(1 + \cos \theta)$ 的全长, 其中 $a > 0$ 是常数.



$$\begin{aligned}
 S &= 2 \int_0^{\pi} \sqrt{r^2 + (r')^2} d\theta = 2 \int_0^{\pi} \sqrt{a^2(1+\cos\theta+2\cos\theta) + a^2\sin^2\theta} d\theta \\
 &= 2 \int_0^{\pi} \sqrt{2a^2(1+\cos\theta)} d\theta \\
 &= 2\sqrt{2}a \int_0^{\pi} \sqrt{1+\cos\theta} d\theta \\
 &= 2\sqrt{2}a \int_0^{\pi} \sqrt{1+\cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}} d\theta \\
 &= 4a \int_0^{\pi} \cos\frac{\theta}{2} d\frac{\theta}{2} = 8a \sin\frac{\theta}{2} \Big|_0^{\pi} = 8a
 \end{aligned}$$

例 15.9

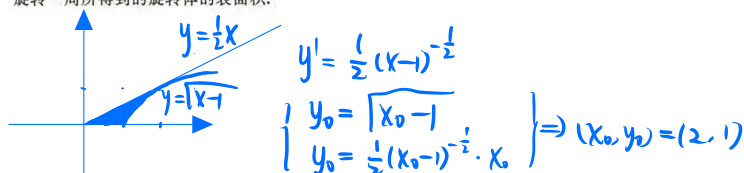
例 15.9 求星形线 $x = \cos^3 t, y = \sin^3 t (0 \leq t \leq 2\pi)$ 的弧长.

$$\begin{aligned}
 S &= 4 \int_0^{\frac{\pi}{2}} \sqrt{(x'_t)^2 + (y'_t)^2} dt \\
 &= 4 \int_0^{\frac{\pi}{2}} \sqrt{[3\cos^2 t \cdot (-\sin t)]^2 + [3\sin^2 t \cos t]^2} dt \\
 &= 12 \int_0^{\frac{\pi}{2}} \sqrt{\cos^2 t \sin^2 t} dt \\
 &= 3 \int_0^{\frac{\pi}{2}} \sin 2t d2t \\
 &= 3(-\cos 2t) \Big|_0^{\frac{\pi}{2}} = 3(1+1) = 6
 \end{aligned}$$

例 15.10

例 15.10 设有曲线 $y = \sqrt{x-1}$, 过原点作其切线, 求由此曲线、切线及 x 轴围成的平面图形绕 x 轴

旋转一周所得到的旋转体的表面积.



$$\begin{aligned}
 S_1 &= 2\pi \int_1^2 \sqrt{x-1} \cdot \sqrt{1 + \left(\frac{1}{2\sqrt{x-1}}\right)^2} dx = 2\pi \int_1^2 \sqrt{x - \frac{3}{4}} dx \\
 &= 2\pi \cdot \frac{2}{3} \left(x - \frac{3}{4}\right)^{\frac{3}{2}} \Big|_1^2 = \frac{4\pi}{3} \left(\frac{\sqrt{5}-1}{8}\right) = \frac{\pi}{6}(\sqrt{5}-1) \\
 S_2 &= 2\pi \int_0^2 \frac{1}{2}x \sqrt{1 + \frac{1}{4}} dx = \frac{\sqrt{5}}{2}\pi \int_0^2 x dx = \sqrt{5}\pi \\
 S &= S_1 + S_2
 \end{aligned}$$

三、微分方程的物理应用

主要涉及以下两个方面的问题.

(1) 牛顿第二定律.

相关物理量有①物体质量 m ; ②力 f (包括重力、阻力、浮力); ③加速度 $a = \frac{d^2x}{dt^2} = \frac{dv}{dt} = \frac{dv}{dx} \frac{dx}{dt} = v \frac{dv}{dx}$. 见

例 15.11.

(2) 变化率问题.

提法多为“ t 时刻某量 y 对 t 的变化率与 t 时刻某量成正比”, 如

① 考题 1 (事实上此题叫冷却定律).

“ t 时刻物体温度 $T(t)$ 对时间的变化率与 t 时刻物体和介质的温差 $T - T_0$ 成正比 ($T > T_0$)”, 应写成

$\frac{dT}{dt} = -k(T - T_0)$, 负号代表 $\frac{dT}{dt} < 0$ (温度随着时间的增加而降低). 见例 15.12.

② 考题 2.

设总人数为 N , “ t 时刻已掌握新技术的人数 x 的变化率和已掌握新技术与未掌握新技术的人数之积成正比”, 应写成 $\frac{dx}{dt} = +kx(N - x)$, 正号代表 $\frac{dx}{dt} > 0$ (人数随着时间增加而增多). 见习题 15.3.

例 15.11

例 15.11 某种飞机在机场降落时, 为了减少滑行距离, 在触地的瞬间, 飞机尾部张开减速伞, 以增大阻力, 使飞机迅速减速并停下.

现有一质量为 9000 kg 的飞机, 着陆时的水平速度为 700 km/h . 经测试, 减速伞打开后, 飞机所受的总阻力与飞机的速度成正比 (比例系数为 $k = 6.0 \times 10^6$). 问从着陆点算起, 飞机滑行的最长距离是多少? (kg 表示千克, km/h 表示千米/时)

$$F = ma = -kV$$

$$m \cdot \frac{dv}{dx} V = -kV$$

$$9000 \cdot V'_x = -6 \times 10^6 \Rightarrow V'_x = -\frac{2}{3} \times 10^3$$

$$\begin{cases} V(x) = -\frac{2}{3} \times 10^3 \cdot x + C \\ V(0) = 700 \end{cases} \Rightarrow C = 700$$

$$\Rightarrow V(x) = -\frac{2}{3} \times 10^3 x + 700$$

$$V = 0 \text{ 时 } x = \frac{700 \times \frac{3}{2}}{1000} = 1.05 \text{ km}$$

例15.12

例 15.12 已知高温物体置于低温介质中,任一时刻该物体温度对时间的变化率与该时刻物体和介质的温差成正比. 现将一初始温度为 120°C 的物体在 20°C 恒温介质中冷却, 30 min 后该物体温度降至 30°C , 若要将该物体的温度继续降至 21°C , 还需冷却多长时间?

$$\begin{aligned}\frac{dT}{dt} &= -k(T-20) \\ \int \frac{dT}{T-20} &= \int -k dt \\ \ln|T-20| &= -kt + C \\ \Rightarrow \ln 100 &= -k \cdot 0 + C \Rightarrow C = \ln 100 \\ \ln 10 &= -30k + \ln 100 \\ k &= \frac{\ln 10}{30} \\ \ln 1 = 0 &= -\frac{\ln 10}{30} t + \ln 100 \\ \Rightarrow t &= 60 \\ \Rightarrow 60 - 30 &= 30\end{aligned}$$

欧拉方程

$$x^2 \frac{d^2 y}{dx^2} + px \frac{dy}{dx} = f(x)$$

$\Downarrow$ = 阶变系数
换元 $\Downarrow$ = 阶常系数

$$\text{方法: } \frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx}$$

形如 $x^2 \frac{d^2 y}{dx^2} + px \frac{dy}{dx} + qy = f(x)$ 的方程称为欧拉方程, 其中 p 与 q 为常数, $f(x)$ 为已知连续函数. 它有固定的解法.

(1) 当 $x > 0$ 时, 令 $x = e^t$, 则 $t = \ln x$, $\frac{dt}{dx} = \frac{1}{x}$, 于是

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{1}{x} \frac{dy}{dt}, \quad \frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{1}{x} \frac{dy}{dt} \right) = -\frac{1}{x^2} \frac{dy}{dt} + \frac{1}{x} \frac{d}{dx} \left(\frac{dy}{dt} \right) = -\frac{1}{x^2} \frac{dy}{dt} + \frac{1}{x^2} \frac{d^2 y}{dt^2},$$

方程化为

$$\frac{d^2 y}{dt^2} + (p-1) \frac{dy}{dt} + qy = f(e^t),$$

即可求解(别忘了用 $t = \ln x$ 回代成 x 的函数).

(2) 当 $x < 0$ 时, 令 $x = -e^t$, 同理可得.

例15.13

例 15.13 欧拉方程 $x^2 \frac{d^2 y}{dx^2} + 4x \frac{dy}{dx} + 2y = 0 (x > 0)$ 的通解为_____.

$$\begin{aligned}x = e^t \quad \frac{d^2 y}{dt^2} + 3 \frac{dy}{dt} + 2y &= 0 \\ \lambda^2 + 3\lambda + 2 &= (\lambda+2)(\lambda+1) = 0\end{aligned}$$

$$\begin{aligned}y(t) &= C_1 e^{-2t} + C_2 e^{-t} \\ y(x) &= C_1 e^{-2 \ln t} + C_2 e^{-\ln t} \\ &= C_1 t^{-2} + C_2 t^{-1}\end{aligned}$$

傅里叶级数

$$f(x) \sim S(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi}{l} x + b_n \sin \frac{n\pi}{l} x \right)$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cdot \cos \frac{n\pi}{l} x \, dx, \quad n=0,1,2,\dots$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \cdot \sin \frac{n\pi}{l} x \, dx, \quad n=1,2,3,\dots$$

$$[-l, l] \begin{cases} \textcircled{1} f(x) \text{ 普通: 直接带} \\ \textcircled{2} f(x) \text{ 奇: } a_n=0, \quad b_n = \frac{2}{l} \int_0^l f(x) \cdot \sin \frac{n\pi}{l} x \, dx \\ \quad f(x) \sim S(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi}{l} x \quad [\text{正弦级数}] \\ \textcircled{3} f(x) \text{ 偶: } b_n=0, \quad a_n = \frac{2}{l} \int_0^l f(x) \cdot \cos \frac{n\pi}{l} x \, dx \\ \quad f(x) \sim S(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi}{l} x \quad [\text{余弦级数}] \end{cases}$$

$$[0, l] \begin{cases} \textcircled{4} \text{ 奇延拓} \\ \text{偶延拓} \end{cases}$$

$$\textcircled{5} f(x) \sim S(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi}{l} x + b_n \sin \frac{n\pi}{l} x$$

$$S(x) = \begin{cases} f(x), & x \text{ 为连续点} \\ \frac{f(x-0) + f(x+0)}{2}, & x \text{ 为间断点} \\ \frac{f(-l+0) + f(l+0)}{2}, & x = \pm l \end{cases}$$

例 15.15

例 15.15 将函数 $f(x) = 1 - x^2 (0 \leq x \leq \pi)$ 展开成余弦级数, 并求级数 $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$.

余弦级数 $\Rightarrow$ 偶延拓

$$f(x) = \begin{cases} 1-x^2 \\ 1-(-x)^2 \end{cases} = 1-x^2 \quad (-\pi \leq x \leq \pi)$$

$$\begin{aligned} f(x) \sim S(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi}{l} x + b_n \sin \frac{n\pi}{l} x \\ &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi}{l} x \end{aligned}$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) \, dx = \frac{2}{\pi} \left[x - \frac{x^3}{3} \right]_0^{\pi} = 2 - \frac{2}{3}\pi^2$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos \frac{n\pi}{l} x \, dx = \frac{2}{\pi} \int_0^{\pi} (1-x^2) \cos nx \, dx$$

$$= \frac{2}{\pi} \int_0^{\pi} \cos nx \, dx - \frac{2}{\pi} \int_0^{\pi} x^2 \cos nx \, dx$$

$$= -\frac{2}{\pi} \int_0^{\pi} x^2 \sin nx \, dx \quad \cos n\pi = (-1)^n$$

$$= -\frac{2}{\pi} \left[2x \frac{\cos nx}{n^2} \right]_0^{\pi} = -\frac{2}{\pi} \cdot \frac{2\pi}{n^2} (-1)^n = -\frac{4}{n^2} (-1)^n$$

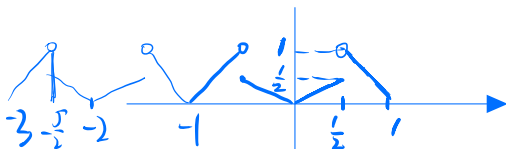
$$S(x) = \left(2 - \frac{2}{3}\pi^2\right) + \sum_{n=1}^{\infty} \frac{4}{n^2} (-1)^{n+1} \cos nx$$

$$\begin{array}{ccccccc} x^2 & \oplus & 2x & \ominus & 2 & \oplus & 0 \\ \cos nx & \frac{\sin nx}{n} & -\frac{\cos nx}{n^2} & \frac{\sin nx}{n^3} & & & \end{array}$$

例 15.14

例 15.14 设 $f(x) = \begin{cases} x, & 0 \leq x \leq \frac{1}{2}, \\ 2-2x, & \frac{1}{2} < x \leq 1, \end{cases}$ $S(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos n\pi x, -\infty < x < +\infty$, 其中 $a_n = 2 \int_0^1 f(x) \cos n\pi x dx (n = 0, 1, 2, \dots)$. 求 $S(-\frac{5}{2})$.

$$S(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos n\pi x \Rightarrow \text{偶延拓}$$



$$S(-\frac{5}{2}) = S(\frac{1}{2}) = \frac{\frac{1}{2} + 1}{2} = \frac{3}{4}$$

例 15.15

例 15.15 将函数 $f(x) = 1 - x^2 (0 \leq x \leq \pi)$ 展开成余弦级数, 并求级数 $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$.

$$f(x) \sim S(x) = (2 - \frac{2}{3}\pi^2) + 4 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} \cdot \cos nx$$

$$\text{令 } x=0, \quad S(0) = (2 - \frac{2}{3}\pi^2) + 4 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = f(0) = 1$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}$$

习 15.1

15.1 甲车以 24 km/h 的速度向北行驶, 同时正东 10 km 处有乙车以 20 km/h 的速度向东行驶. 从这一时刻起经过 1 小时后, 求两车间的距离对时间的变化率.

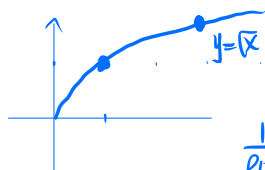


$$y = \sqrt{(24t)^2 + (10 + 20t)^2}$$

$$y' = \frac{48t \cdot 24 + 2(10 + 20t) \cdot 20}{2 \sqrt{(24t)^2 + (10 + 20t)^2}} \Big|_{t=1} = \frac{24^2 + 600}{\sqrt{24^2 + 900}}$$

习 15.2

15.2 设 $\rho = \rho(x)$ 是抛物线 $y = \sqrt{x}$ 上任一点 $M(x, y) (x \geq 1)$ 处的曲率半径, $s = s(x)$ 是该抛物线上介于点 $A(1, 1)$ 与 M 之间的弧长, 计算 $3\rho \frac{d^2\rho}{ds^2} - (\frac{d\rho}{ds})^2$ 的值.



$$y' = \frac{1}{2}x^{-\frac{1}{2}}$$

$$y'' = -\frac{1}{4}x^{-\frac{3}{2}}$$

$$\frac{1}{\rho(x)} = k = \frac{|y''|}{(1 + (y')^2)^{\frac{3}{2}}} = \frac{\frac{1}{4}x^{-\frac{3}{2}}}{(1 + \frac{1}{4x})^{\frac{3}{2}}} = \frac{1}{4} \left(\frac{\frac{1}{x}}{1 + \frac{1}{4x}} \right)^{\frac{3}{2}} = \frac{1}{4} \left(\frac{4}{4x+1} \right)^{\frac{3}{2}}$$

$$\rho(x) = \frac{(4x+1)^{\frac{3}{2}}}{2} \quad \rho'_x = \frac{3}{4} (4x+1)^{\frac{1}{2}} \cdot 4 = 3(4x+1)^{\frac{1}{2}}$$

$$S = \int_1^x \sqrt{1+(y')^2} dt = \int_1^x \sqrt{1+\frac{1}{4t^2}} dt = \int_1^x \sqrt{\frac{4t+1}{4t}} dt$$

$$3p \frac{d^2 p}{ds^2} - \left(\frac{dp}{ds} \right)^2$$

$$\frac{dp}{ds} = \frac{dp}{dx} \cdot \frac{dx}{ds} = 3(4x+1)^{\frac{1}{2}} \cdot \sqrt{\frac{4x}{4x+1}} = 6\sqrt{x}$$

$$\frac{d^2 p}{ds^2} = \frac{d \frac{dp}{ds}}{ds} = \frac{d \frac{dp}{dx}}{dx} \cdot \frac{dx}{ds} = 3 \cdot \frac{1}{\sqrt{x}} \cdot \sqrt{\frac{4x}{4x+1}} = \frac{6}{\sqrt{4x+1}}$$

$$3p \frac{d^2 p}{ds^2} - \left(\frac{dp}{ds} \right)^2 = \frac{3}{2} (4x+1)^{\frac{3}{2}} \cdot \frac{6}{\sqrt{4x+1}} - 36x$$

$$= 9(4x+1) - 36x = 9$$

例15.3

15.3 在某一人群中推广新技术是通过其中已掌握新技术的人进行的. 设该人群的总人数为 N , 在 $t=0$ 时刻已掌握新技术的人数为 x_0 , 在任意时刻 t 已掌握新技术的人数为 $x(t)$ (将 $x(t)$ 视为连续可微变量), 其变化率与已掌握新技术人数和未掌握新技术人数之积成正比, 比例常数 $k > 0$, 求 $x(t)$.

$$\frac{x'_t}{x(N-x)} = k$$

$$\frac{dx}{dt} = k(x(N-x))$$

$$\int \frac{1}{x(N-x)} dx = \int k dt$$

$$\frac{1}{N} \int \frac{1}{x} + \frac{1}{N-x} dx = kt$$

$$\ln x(N-x) C = Nkt$$

$$e^{Nkt} = C x(N-x)$$

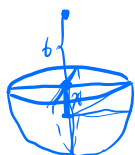
$$t=0 \text{ 时 } 1 = C \cdot x_0 \cdot (N-x_0)$$

$$\Rightarrow C = \frac{1}{x_0(N-x_0)}$$

$$x(N-x) = x_0(N-x_0) e^{Nkt}$$

例15.4

15.4 有一半径为 4 m 的半球形水池蓄满了水, 现在要将水全部抽到距水池原水面 6 m 高的水箱中, 求需做多少功 (水的密度为 1000 kg/m^3 , 重力加速度 $g=9.8 \text{ m/s}^2$, $\pi=3.14$).



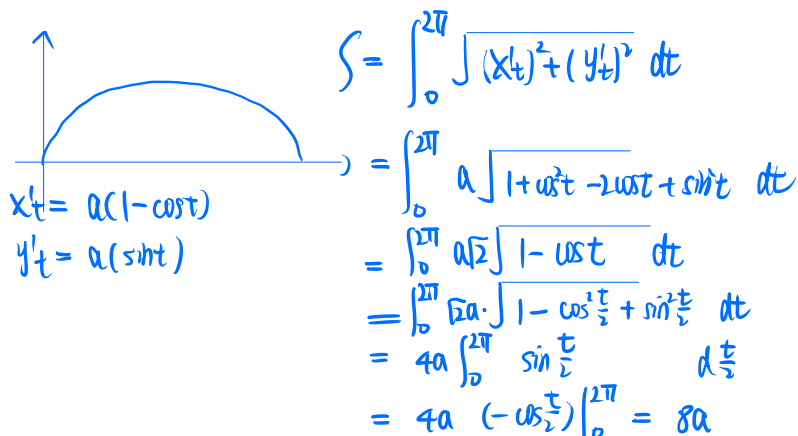
$$W = \int_0^4 \rho g (6+x) \cdot \pi (16-x^2) dx$$

$$= \rho g \pi \int_0^4 (-x^3 - 6x^2 + 16x + 96) dx$$

$$= \rho g \pi \left[-\frac{x^4}{4} - 2x^3 + 8x^2 + 96x \right]_0^4 = \rho g \pi \cdot 320$$

习 15.5

15.5 求摆线 $\begin{cases} x=a(t-\sin t), \\ y=a(1-\cos t) \end{cases}$ ($a>0$) 的一拱 ($0\leq t\leq 2\pi$) 的弧长.



$$\begin{aligned}
 S &= \int_0^{2\pi} \sqrt{(x'_t)^2 + (y'_t)^2} dt \\
 &= \int_0^{2\pi} a \sqrt{1 + \cos^2 t - 2\cos t + \sin^2 t} dt \\
 &= \int_0^{2\pi} a \sqrt{2 - 2\cos t} dt \\
 &= \int_0^{2\pi} \sqrt{2} a \sqrt{1 - \cos t} dt \\
 &= \int_0^{2\pi} \sqrt{2} a \sqrt{1 - \cos^2 \frac{t}{2} + \sin^2 \frac{t}{2}} dt \\
 &= 4a \int_0^{2\pi} \sin \frac{t}{2} d\frac{t}{2} \\
 &= 4a (-\cos \frac{t}{2}) \Big|_0^{2\pi} = 8a
 \end{aligned}$$

习 15.6

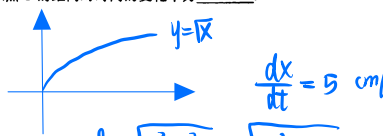
求阿基米德螺线 $r=a\theta$ ($a>0$) 上相应于 θ 从 0 到 2π 一段的弧长.



$$\begin{aligned}
 S &= \int_0^{2\pi} \sqrt{r^2 + (r')^2} d\theta \\
 &= \int_0^{2\pi} \sqrt{a^2\theta^2 + a^2} d\theta = \int_0^{2\pi} a \sqrt{\theta^2 + 1} d\theta \\
 \theta = \tan t &= a \int_0^{\arctan 2\pi} \sec t d\tan t = a \left[\sec t \cdot \tan t \Big|_0^{\arctan 2\pi} - \int_0^{\arctan 2\pi} \tan^2 t \cdot \sec t dt \right] \\
 &= a \left[2\pi \sqrt{4\pi^2 + 1} - \int_0^{\arctan 2\pi} \sec^3 t dt + \int_0^{\arctan 2\pi} \sec t dt \right] \\
 \Rightarrow S &= \frac{a}{2} \left[2\pi \sqrt{4\pi^2 + 1} + \ln |\sec t + \tan t| \Big|_0^{\arctan 2\pi} \right] \\
 &= \frac{a}{2} \left[2\pi \sqrt{4\pi^2 + 1} + \ln(2\pi + \sqrt{4\pi^2 + 1}) \right]
 \end{aligned}$$

例 15.1

15.1 质点 P 沿抛物线 $x=y^2$ ($y>0$) 移动, P 的横坐标 x 对时间的变化率为 5 cm/s. 当 $x=9$ 时, 点 P 到原点 O 的距离对时间的变化率为_____.



$$\begin{aligned}
 l &= \sqrt{x^2 + y^2} = \sqrt{x^2 + x} \\
 \frac{dl}{dt} &= \frac{dl}{dx} \frac{dx}{dt} = \frac{2x+1}{2\sqrt{x^2+x}} \cdot 5 \Big|_{x=9} = \frac{19 \times 5}{2 \times 3 \sqrt{10}} = \frac{19\sqrt{10}}{120}
 \end{aligned}$$

300.15.2

15.2 设 $y=y(x)$ 由参数方程 $\begin{cases} x=1+t^2 \\ y=\cos t \end{cases}$ 所确定, 求曲线 $y=y(x)$ 在 $t=\frac{\pi}{2}$ 对应点处的切线方程.

$$k = \frac{|y'|}{(1+(y')^2)^{\frac{1}{2}}} \Big|_{t=\frac{\pi}{2}} = \frac{\frac{2}{\pi^3}}{(1+\frac{1}{\pi^2})^{\frac{1}{2}}} = \frac{\frac{2}{\pi^3}}{\frac{\pi^2+1}{\pi^3}} = \frac{2}{(\pi^2+1)^{\frac{1}{2}}}$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-\sin t}{2t} \Big|_{t=\frac{\pi}{2}} = \frac{-1}{\pi} \quad \checkmark$$

$$\frac{d \frac{dy}{dx}}{dk} = \frac{\frac{-\cos t \cdot 2t + \sin t \cdot 2}{4t^3}}{\frac{2}{8t^3}} = \frac{2\sin t - 2t \cdot \cos t}{8t^3} \Big|_{t=\frac{\pi}{2}} = \frac{2}{\pi^3}$$

$$x|_{t=\frac{\pi}{2}} = 1 + \frac{\pi^2}{4}$$

$$y|_{t=\frac{\pi}{2}} = 0$$

$$y = \frac{2}{(\pi^2+1)^{\frac{1}{2}}} (x - 1 - \frac{\pi^2}{4})$$

300.15.3

设 $y=y(x)$ 由参数方程 $\begin{cases} x=t+e^t \\ y=\sin t \end{cases}$ 确定, 则 $\frac{d^2y}{dx^2} \Big|_{t=0} = \underline{\quad\quad}$.

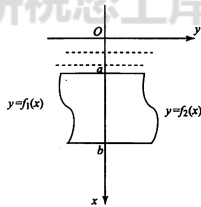
$$\frac{dy}{dx} = \frac{\cos t}{1+e^t}$$

$$\frac{d^2y}{dx^2} = \frac{\frac{-\sin t(1+e^t) - \cos t \cdot e^t}{(1+e^t)^2}}{1+e^t} \Big|_{t=0} = \frac{-1}{2}$$

300.15.4

15.4 由曲线 $y=f_1(x)$, $y=f_2(x)$ 及直线 $x=a$, $x=b$ ($a < b$) 所围成的平面板铅直地投入容重为 r ($r = \rho g$ 表示单位体积液体的重力) 的液体中, x 轴铅直向下, 液面与 y 轴重合, 如图所示, 平面板所受液压力为

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$$F = \int_a^b \rho g \cdot x (f_2(x) - f_1(x)) dx \quad (B) \quad F = \int_a^b \rho g \cdot x |f_2(x) - f_1(x)| dx \quad (D)$$

300.15.5

求抛物线 $6y=x^2$ 上从点 $(0,0)$ 到点 $(4, \frac{8}{3})$ 之间的弧长.

$$S = \int_0^4 \sqrt{1 + (\frac{x}{3})^2} dx = 3 \int_0^{\frac{4}{3}} \sqrt{1+t^2} dt = 3 \left[\frac{1}{2} t \sqrt{1+t^2} + \frac{1}{2} \ln |t + \sqrt{1+t^2}| \right] \Big|_0^{\frac{4}{3}} = \frac{10}{3} + \frac{3}{2} \ln \frac{8}{3}$$

300.15.6

15.6 设某物体的温度 T 与时间 t 满足函数关系:

$$T = a(1 - e^{-kt}) + b,$$

其中 T 的单位是 $^{\circ}\text{C}$, t 的单位是 min. 现将该物体放入 200°C 的高温介质中.

(1) 若物体的初始温度是 20°C , 求 a 和 b ;

(2) 在(1)的条件下, 若物体温度以 $2^{\circ}\text{C}/\text{min}$ 的速率开始上升, 求 k .

$$(1) \quad 20 = a(1 - e^0) + b \Rightarrow b = 20$$

$$\lim_{t \rightarrow +\infty} T = a + b = 200 \Rightarrow a = 180$$

$$\Rightarrow T = 180(1 - e^{-kt}) + 20$$

$$(2) \quad \frac{dT}{dt} = 180 \cdot k e^{-kt} = 2$$

$$k = \frac{1}{90}$$

300.15.7

15.7(仅数学一) 求方程 $x^2 \frac{d^2 y}{dx^2} - 2y = x^2$ 的通解.

欧拉方程

$$\textcircled{1} \quad \text{设 } x = e^t$$

$$\textcircled{2} \Rightarrow y_{tt} - 2y = e^{2t}$$

$$\lambda^2 - 2 = 0 \Rightarrow \lambda = \pm\sqrt{2}$$

$$y = C_1 e^{\sqrt{2}t} + C_2 e^{-\sqrt{2}t} = C_1 x^{\sqrt{2}} + C_2 x^{-\sqrt{2}}$$

$$\textcircled{3} \quad \text{设 } y^* = e^{2t} \cdot A \cdot x^0 = e^{2t} \cdot A$$

$$\therefore 4A \cdot e^{2t} - 2A e^{2t} = e^{2t} \Rightarrow A = \frac{1}{2}$$

$$\therefore y^* = \frac{1}{2} e^{2t} = \frac{1}{2} x^2$$

$$\textcircled{4} \quad y = \frac{1}{2} x^2 + C_1 x^{\sqrt{2}} + C_2 x^{-\sqrt{2}}$$

$$\textcircled{1} \text{ 设 } x = e^t$$

$$\textcircled{2} \quad y_{tt} - y_t' - 2y = e^{2t}$$

$$\lambda^2 - \lambda - 2 = (\lambda - 2)(\lambda + 1) = 0 \Rightarrow \lambda_1 = 2, \lambda_2 = -1$$

$$y = C_1 e^{2t} + C_2 e^{-t} = C_1 x^2 + C_2 x^{-1}$$

$$\textcircled{3} \quad y^* = e^{2t} \cdot A \cdot t = A t \cdot e^{2t}$$

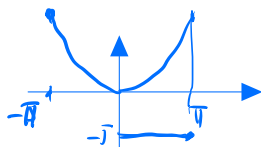
$$4A e^{2t}(1+t) - (A e^{2t} + 2A t e^{2t}) - 2A t \cdot e^{2t} = e^{2t}$$

$$\Rightarrow y^* = \frac{1}{3} \ln x \cdot x^2 \Rightarrow A = \frac{1}{3}$$

$$\textcircled{4} \quad y = \frac{1}{3} x^2 \ln x + C_1 x^2 + C_2 x^{-1}$$

300.15.8

15.8(仅数学一) 设 $f(x) = \begin{cases} x^2, & -\pi \leq x < 0, \\ -5, & 0 \leq x < \pi, \end{cases}$ 则其以 2π 为周期的傅里叶级数在 $x = \pm\pi$ 处收敛于



$$-\frac{5 + \pi^2}{2}$$

300.15.9

15.9 (仅数学一) 函数 $f(x) = \begin{cases} -1, & -\pi \leq x < 0, \\ 1, & 0 \leq x \leq \pi \end{cases}$ 在 $[-\pi, \pi]$ 上展开为傅里叶级数

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx),$$

则 $a_n = 0$, $b_n =$ _____, 和函数 $S(x) =$ _____.

$$b_n = \frac{2}{\pi} \int_0^{\pi} \sin nx \, dx = \frac{2}{\pi} \int_0^{\pi} \sin nx \, dx = \frac{-2}{n\pi} \cos nx \Big|_0^{\pi} = \frac{-2}{n\pi} [(-1)^n - 1]$$

$$S(x) = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n} \sin nx$$

300.15.10

15.10 (仅数学一) 将函数 $f(x) = 2 + |x|$ ($-1 \leq x \leq 1$) 展开成以 2 为周期的傅里叶级数, 并由此求级数 $\sum_{n=1}^{\infty} \frac{1}{n^2}$ 的和.偶函数 $\Rightarrow b_n = 0$

$$a_0 = 2 \int_0^1 f(x) \, dx = 2 \int_0^1 (2+x) \, dx = 2 \cdot \frac{5}{2} = 5$$

$$a_n = 2 \int_0^1 (2+x) \cos n\pi x \, dx$$

$$= 4 \int_0^1 \cos n\pi x \, dx + 2 \int_0^1 x \cdot \cos n\pi x \, dx$$

$$= x \cdot \frac{\sin n\pi x}{n\pi} + \frac{2 \cos n\pi x}{n^2 \pi^2} \Big|_0^1$$

$$= 2 \frac{\cos n\pi - 1}{n^2 \pi^2} = 2 \frac{(-1)^n - 1}{n^2 \pi^2}$$

$$f(x) \sim S(x) = \frac{5}{2} + \frac{2}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^2} \cdot \cos n\pi x$$

$$2 = f(0) = S(0) = \frac{5}{2} + \frac{2}{\pi^2} \left(\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} - \frac{1}{n^2} \right)$$

$$= \frac{5}{2} + \frac{2}{\pi^2} \left[-2 + 0 + \frac{-2}{3^2} + 0 + \frac{-2}{5^2} + \dots \right]$$

$$\ominus \frac{5}{2} \quad 2 = \frac{5}{2} \ominus \frac{4}{\pi^2} \left(\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} \right)$$

$$\Rightarrow \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \frac{\pi^2}{8}$$

$$\left(\sum_{n=1}^{\infty} \frac{1}{n^2} \right) = \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} + \sum_{n=1}^{\infty} \frac{1}{(2n)^2} = \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} + \frac{1}{4} \sum_{n=1}^{\infty} \frac{1}{n^2}$$

$$\therefore X = \frac{\pi^2}{8} + \frac{1}{4} X \quad \Rightarrow X = \frac{\pi^2}{6}$$

